Preferential accretion onto eccentric and unequal binary black holes

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ABSTRACT

Supermassive binary black holes (SMBBHs) are expected to be surrounded by circumbinary disks (CBDs) which affect the binary through gravitational forces and accretion. It has been reported that the binary can experience "preferential accretion" where one black hole (BH) out-accretes the other for hundreds of orbits, but this asymmetry has yet to be fully described or understood. In this work, we utilize a suite of 80 SMBBH hydrodynamic simulations with varying mass ratios (q_b) and eccentricity (e_b) in order to robustly delineate the behavior of preferential accretion, determine its relationship to the structure of the CBD, and study its observational consequences. We report measurements of the binary mass-ratio rate-of-change $\dot{q} \equiv d/dt(M_2/M_1)$ and the accretion-rate ratio $\lambda(t) \equiv \dot{M}_2(t)/\dot{M}_1(t)$, both as functions of q_b and e_b. We also find that a) the time-variability of the accretion-rate ratio onto the two BHs tracks the precession of the CBD, b) there can be sub- and super-Eddington accretion in a single binary, and c) there exists a regime where the binary evolves toward $q_b \neq 1$ during the gas-dominated phase. Given these findings, we suggest flickering jets and an under-density of LISA events with $q_b = 1$ as potential observable consequences.

1 INTRODUCTION

Cosmic structure forms hierarchically and galaxy mergers are expected to result in gravitationally bound supermassive binary black holes (SMBBHs) (White & Rees 1978; Begelman et al. 1980); the same disc-mediated dynamics also operate in binary stars and protoplanetary systems. During galactic merger, it is also expected that the inter-stellar medium from the proto-galaxies are funneled to the galactic center (Barnes & Hernquist 1992). This creates a reservoir of material that, due to the conservation of angular momentum and the binary's gravitational potential, forms a circumbinary disk (CBD). Many aspects of the CBD have been well studied in the literature, in both the SMBBH and the star-planet contexts. It has been shown that the binary can influence the CBD, creating eccentric structure and precessing eigenmodes, (Lubow 1991; Whitehurst 1994; Nelson 2003; Goodchild & Ogilvie 2006; Paardekooper et al. 2008; Kley et al. 2008; Shi et al. 2012; Miranda et al. 2017; Thun et al. 2017; Muñoz & Lithwick 2020; Lubow 2022; Siwek et al. 2023a) and that the CBD can in turn influence the binary—changing both its semi-major axis and eccentricity (Tiede et al. 2020; Zrake et al. 2021; D'Orazio et al. 2013; Siwek et al. 2023b; D'Orazio & Duffell 2021; Farris et al. 2014; Muñoz et al. 2019), as well as exciting precession of the binary itself (Tiede et al. 2024; Dittmann et al. 2023; Calcino et al. 2023). However, crucially, the binary also accretes from the circumbinary disk, influencing both the mass-ratio and associated light-curves of the system (e.g Siwek et al. 2023a; Westernacher-Schneider et al. 2022; Farris et al. 2014; D'Orazio et al. 2013).

It has been a widely known result of CBD studies that the accretion on to the binary ($\dot{M}_\mathrm{b} = \dot{M}_1 + \dot{M}_2$) is not steady but experiences variability through time. Near equal-mass ratio ($q_b \geq 0.7$) circular binaries display a sawtooth pattern with a period of about 5 binary orbital periods (τ_b) potentially due to the presence of an m = 1 over-density traveling at the inner edge of the CBD (MacFadyen & Milosavljević 2008; Miranda et al. 2017; Muñoz et al. 2019; Westernacher-Schneider et al. 2022). Eccentric and unequal mass-ratio binaries also display variability on the order of a binary period (Miranda et al. 2017; Muñoz et al. 2019; Westernacher-Schneider et al. 2022). In order to study how the binary accretes material from the circumbinary disk, Tiede et al. (2022) tracked the paths of passive tracer particles in a two-dimensional grid-based hydrodynamical code (DISCO). They determined that much of the gas is viscously transported from the outer disk before adopting a ballistic trajectory and ultimately accreting on to the BHs near the inner edge. They also delineate an "accretion horizon" at $r \approx 1.01\, a_b$ (the binary semi-major axis), a radius past which any material entering is accreted.

However, studies have also pointed out the discrepancy between the accretion rates of the primary and secondary components of the SMBBH (e.g. Miranda et al. 2017; Siwek et al. 2023a; D'Orazio et al. 2024). Namely, the secondary tends to accrete, on average, at a greater rate than the primary in what has become known as "preferential accretion" — pushing the system toward equal mass ratio ($q_b = 1$; Farris et al. 2014; Duffell et al. 2020; Miranda et al. 2017; Siwek et al. 2023a). While systems with $q_b = 1$ generally accrete at equal rates, suggesting a stable equilibrium at $q_b = 1$, studies have reported that at certain eccentricities (e.g. $e_b = 0.5$ and $e_b = 0.6$) the binary can undergo transient "symmetry breaking" (see Figure 7 of Muñoz et al. 2019). In these instances, one black hole temporarily accretes more than its counterpart before its accretion rate is suppressed, allowing the other to experience an enhanced accretion-rate episode.

This process was explored in Siwek et al. 2023a, hereafter S23, by studying the accretion rates across 50 2D hydrodynamic simulations of SMBBHs in CBDs, sampling different binary parameters q_b and e_b. S23 reported time averaged values of $\lambda = \dot{M}_2/\dot{M}_1$, the ratio of the secondary's accretion rate to the primary's, for their simulation suite and displayed snapshots of the corresponding CBD. S23 found that low-eccentricity binaries display larger values of λ than high-eccentricity binaries ($e_b > 0.6$). Further, the process of the flip in preferential accretion was suggested to be associated with the behavior of the CBD. Namely, it was associated with a unique disk behavior they called "forced precession". DeLaurentiis et al. 2025 also discussed the preferential accretion rate of the binary, suggesting that it can be understood in geometric terms, through the distance from the individual BHs to the CBD.

The protoplanetary-disk community has run two- and three-dimensional smoothed-particle hydrodynamic (SPH) simulations of CBDs to understand the accretion onto the primary and secondary (Günther & Kley 2002; Ochi et al. 2005; Young et al. 2015). To date, in simulations run for up to $\sim$ 100 binary orbital periods (τ_b), they have found that the primary can out-accrete the secondary due to streamlines that, despite entering the Lagrange point L2 near the secondary, carry enough angular momentum to flow back to the primary (Ochi et al. 2005; Young et al. 2015). Similar findings are reported by Tiede et al. (2022).

Despite these efforts we are still yet to fully characterize and understand the rich detail of accretion onto the individual components of a binary in a CBD system. Thus, we build on S23 and use the larger simulation suite of Siwek et al. 2023b to report the largest binary parameter sweep study on "preferential accretion" to date. Further, we characterize the accretion behavior by drawing on linear theory of eccentric circumbinary disks and applying a few targeted numerical techniques: Fourier-based period extraction of the accretion rate ratio, comparison against per-BH Eddington thresholds, and a coupled gas-plus-gravitational-wave integration of the binary mass ratio to study its long-term evolution.

In Section 2 we briefly discuss the technical details of the simulations, the key concepts from linear theory we draw upon, and the numerical techniques we employ. In Section 3 we present our findings, detailing the temporal behavior of preferential accretion and mass-ratio evolution and describing how each varies with q_b and e_b. In Section 4 we discuss the consequences of these findings for observations: in particular, jets that periodically switch on and off ("flickering" jets) and a population of SMBBHs with $q_b \neq 1$. In Section 5 we summarize our key findings and discuss next steps.

2 ANALYTIC TOOLS AND NUMERICAL METHODS

In the following section we briefly describe the setup of the Siwek et al. (2023a,b) simulations, the concepts from linear disk theory we employ, and the numerical tools we leverage.

2.1 Simulation setup

We briefly describe the setup of the simulations of interest and refer readers to Siwek et al. (2023a,b) for a more thorough discussion.

Siwek et al. (2023a,b) performed 2D hydrodynamical simulations of binary black holes embedded in a finite, locally isothermal disk with an α-viscosity of $\alpha = 0.1$ and an aspect ratio of $h = 0.1$. The disk is initialized with a power-law surface-density profile and a corresponding temperature profile; the inner edge of the cavity sits at $2a_b$ and the outer edge at $\approx 50a_b$. The binary-disk system spans a computational grid of $300a_b \times 300a_b$ with open boundary conditions, allowing the disk to settle into a quasi-steady state.

The binary is modeled as two sink particles with a mass-ratio $q_b \equiv \frac{M_2}{M_1} \leq 1$ and radii $r_s = 0.03a_b$, moving on a fixed Keplerian orbit of eccentricity e_b. The fraction of gas accreted from each gas cell at each time step by the sink particle is $\gamma_0 \left(1 - \frac{r_{ij}}{r_s}\right)^2$ where r_{ij} is the radial distance from the $j^{\rm th}$ sink particle to the $i^{\rm th}$ gas cell. In addition to mass, the sink also accretes the gas' linear momentum from the gas cells.

The simulations, conducted with the moving-mesh code AREPO (Springel 2010), use Voronoi tessellations to generate a grid of cells and explore a wide parameter space spanning $q_b \in \{0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0\}$ and $e_b \in \{0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.8\}$. The simulations were run for 10 000 binary orbits, with surface-density snapshots recorded at apocenter every 10 orbits. Accretion rates are recorded independently at a much higher cadence, with a time-step of ≈ 0.01 orbits.

2.2 Disk theory

In order to illuminate the relationship between the accretion of the binary and the behavior of the CBD we must first characterize the time-variable behavior and attributes of the disk. Namely, we focus on the eccentricity and precession of the disk.

Gas eccentricity within CBDs is expected to grow through mechanisms such as eccentric Lindblad resonances (ELRs) or spiral shock pumping at the cavity edge (Lubow 1991; Whitehurst 1994; Paardekooper et al. 2008; Kley et al. 2008; Shi et al. 2012). While ELRs and spiral shocks promote eccentricity growth (Lubow 1991; Shi et al. 2012), viscous damping acts to suppress it (Goodchild & Ogilvie 2006). Previous 2D simulations have found steady-state eccentricity profiles, indicating that these competing effects can reach equilibrium (Miranda et al. 2017; Siwek et al. 2023a).

Further, 2D and 3D hydrodynamical simulations have found significant eccentricity near the inner edge of the disk (the cavity), with eccentricity declining outward (MacFadyen & Milosavljević 2008; Miranda et al. 2017; Siwek et al. 2023a; Ragusa et al. 2024). Similar trends have been seen in magneto-hydrodynamic simulations (Shi et al. 2012), suggesting that eccentricity is a robust characteristic of the inner regions of eccentric CBDs.

Additionally, simulations have demonstrated that CBDs can precess (Nelson 2003; Shi et al. 2012; Miranda et al. 2017; Thun et al. 2017; Siwek et al. 2023a), with precession frequencies often attributed to the eigenmodes of a Schrödinger-like equation for eccentricity evolution (Goodchild & Ogilvie 2006; Shi et al. 2012; Teyssandier & Ogilvie 2016; Lee et al. 2019; Muñoz & Lithwick 2020; Lubow 2022).

A robust study of the eccentricity and precession of both the bulk of the disk and the cavity in hydrodynamic simulations was conducted by DeLaurentiis & Rafikov (in preparation). Utilizing the same suite of simulations as this paper, they delineate the shape of the non-linear inner edge of the circumbinary disk, the cavity, by extracting the dominant Fourier modes of the associated isodensity contour.

While alternative explanations have been proposed (Artymowicz 1983), it has widely been assumed that the preferential accretion of the binary is related to the relative closeness of the components to the CBD's inner edge (Rafikov 2016). In order to test this dependence, we utilize the results for the cavity shape and its precession period from DeLaurentiis & Rafikov (in preparation) to build a robust time-dependent geometric picture of the binary in the cavity.

2.3 Numerical techniques

As discussed earlier, in order to highlight the link between accretion and the behavior of the CBD, we are interested in understanding both as time-dependent quantities. The orientation of the CBD is inherently time-dependent due to its apsidal (rather than nodal) precession[1]. Since our simulation runs output snapshots every 10 orbits, the time-series associated with our CBD is constrained to a 10τ cadence.

The accretion rate is likewise time-dependent. For each sink, the simulation tracks the mass accreted Δm at time-step t, with a maximum cadence of 0.01τ. As we are interested in comparing the accretion rate to the behavior of the CBD, we down-sample the high cadence accretion rate data to yield a time-series that matches the 10 orbit cadence of the CBD time-series. We do this by summing the instantaneous mass accreted over non-overlapping 10-orbit windows and dividing it by the window size (of 10 orbits). After we transform the accretion rates for each BH into the lower cadence of 10 orbits, we construct our preferential accretion quantity $\lambda(t) = \frac{\dot{M}_2(t)}{\dot{M}_1(t)}$, the ratio of the accretion rate of the secondary to the primary.

We also report the rate of change of the mass-ratio $\dot{q}(t)$. We define it as

$$\dot{q} = \frac{[1 + q_b(t)][\lambda(q_b) - q_b(t)]}{1 + \lambda(q_b)} \frac{\dot{M}_1 + \dot{M}_2}{M_1 + M_2} \tag{1}$$

where

$$q(t) \equiv \frac{M_2(t)}{M_1(t)} \tag{2}$$

and the mass of each BH

$$M_j(t) = \sum_0^t (M_j(t = 0) + \Delta m(t)) \tag{3}$$

is simply the running sum. Again, we note that all time-variable inputs are first transformed to be of 10-orbit cadence via the aforementioned procedure.

3 RESULTS

In the following sections we report our key results. Namely, in Section 3.1 we detail the behavior of $\lambda(t)$ and report its mean value and, if time variable, its period. In Section 3.1.1, we make some preliminary investigation into the CBD's effect on the accretion behavior. In Section 3.2 we calculate the corresponding rate of change of the mass ratio, and highlight an instance of the binary accreting away from equal mass ($q_b = 1$).

3.1 Preferential accretion

In Fig. 1 we plot our $\lambda(t)$ time-series for the entirety of our simulation suite. The figure is structured such that each panel represents a unique simulation. The rows represent simulations of the same mass-ratio q_b, whereas the columns represent simulations of the same eccentricity e_b. The left-most columns are the least eccentric binaries, and the upper-most rows are the most equal-mass binaries.

A striking feature of Fig. 1 is the broad division of $\lambda(t)$ into either approximately constant or time-varying behavior. Many low-e_b, low-q_b binaries are time-stable. Though some experience a sharp change in behavior in the first 2000 orbits associated with the expected initial disk-instability transient (Moriwaki & Nakagawa 2004), they soon settle to a near-constant value. This time-stable behavior is

1.0	S	V	S	V	V	V	V	V
0.9	S	V	S	S	V	V	V	V
0.8	S	V	S	S	V	V	V	V
0.7	S	V	S	S	V	V	V	V
0.6	S	V	S	S	V	S	V	V
0.5	S	V	S	S	S	S	V	V
0.4	S	V	S	S	S	S	S	V
0.3	S	V	S	S	S	S	S	V
0.2	S	V	S	S	S	S	S	S
0.1	S	S	S	S	S	S	S	S
q_b/e_b	**0.0**	**0.1**	**0.2**	**0.3**	**0.4**	**0.5**	**0.6**	**0.8**

Table 1. Grid showing whether $\lambda(t)$ for a given binary is time-stable (S, Red) or time-varying (V, Blue).

1.0	P	P	L	P	P	P	P	P
0.9	P	P	L	L	P	P	P	P
0.8	P	P	L	L	P	P	P	P
0.7	P	P	L	L	P	P	P	P
0.6	P	P	L	L	P	L	P	P
0.5	P	P	L	L	L	L	P	P
0.4	P	P	L	L	L	L	L	P
0.3	P	P	L	L	L	L	L	P
0.2	P	P	L	L	L	L	L	L
0.1	P	L	L	L	L	L	L	L
q_b/e_b	**0.0**	**0.1**	**0.2**	**0.3**	**0.4**	**0.5**	**0.6**	**0.8**

Table 2. Reproduction of Table 1 from DeLaurentiis & Rafikov (in preparation), showing whether the CBD about a binary of given q_b and e_b is locked (L; red) or precessing (P; blue). The CBD behavior is determined by evaluating the time-series of the eccentricity vector at $a \leq 5a_b$.

exemplified by the $(e_b, q_b) = (0.3, 0.3)$ binary[2]. Others show time-variable behavior, with oscillations in $\lambda(t)$ spanning as much as two orders of magnitude, such as $(e_b, q_b) = (0.6, 1.0)$.

In Table 1 we delineate whether $\lambda(t)$ is time-varying or time-stable via a blue cell with a V or red cell with an S, respectively. It is of particular note how similar Table 1, which depicts the time-variability of $\lambda(t)$, is to Table 2, Table 1 of DeLaurentiis & Rafikov (in preparation), which depicts the precession state of the CBD. For non-circular binaries, time-varying $\lambda(t)$ corresponds to a forced-precessing CBD and time-stable $\lambda(t)$ corresponds to a locked CBD — suggesting that the time-variability of preferential accretion is, in some part, paced by the behavior of the CBD and thereby the cavity. S23 already noted that forced precession in eccentric binaries is associated with strong modulation of the individual accretion rates on the precession timescale (their Section 3.5), invoking symmetry arguments to argue that circular binaries should remain time-stable

[1] While the eccentricity and semi-major axis of the cavity are strictly time-dependent, for most systems we find that the shape of the cavity achieves a steady state and its time-dependence is dominated by its apsidal precession alone.

[2] Some binaries (e.g. $(e_b, q_b) = (0.0, 0.3)$) experience noticeable noise around the constant value, but the dichotomy between time-varying and time-stable $\lambda(t)$ remains clear.

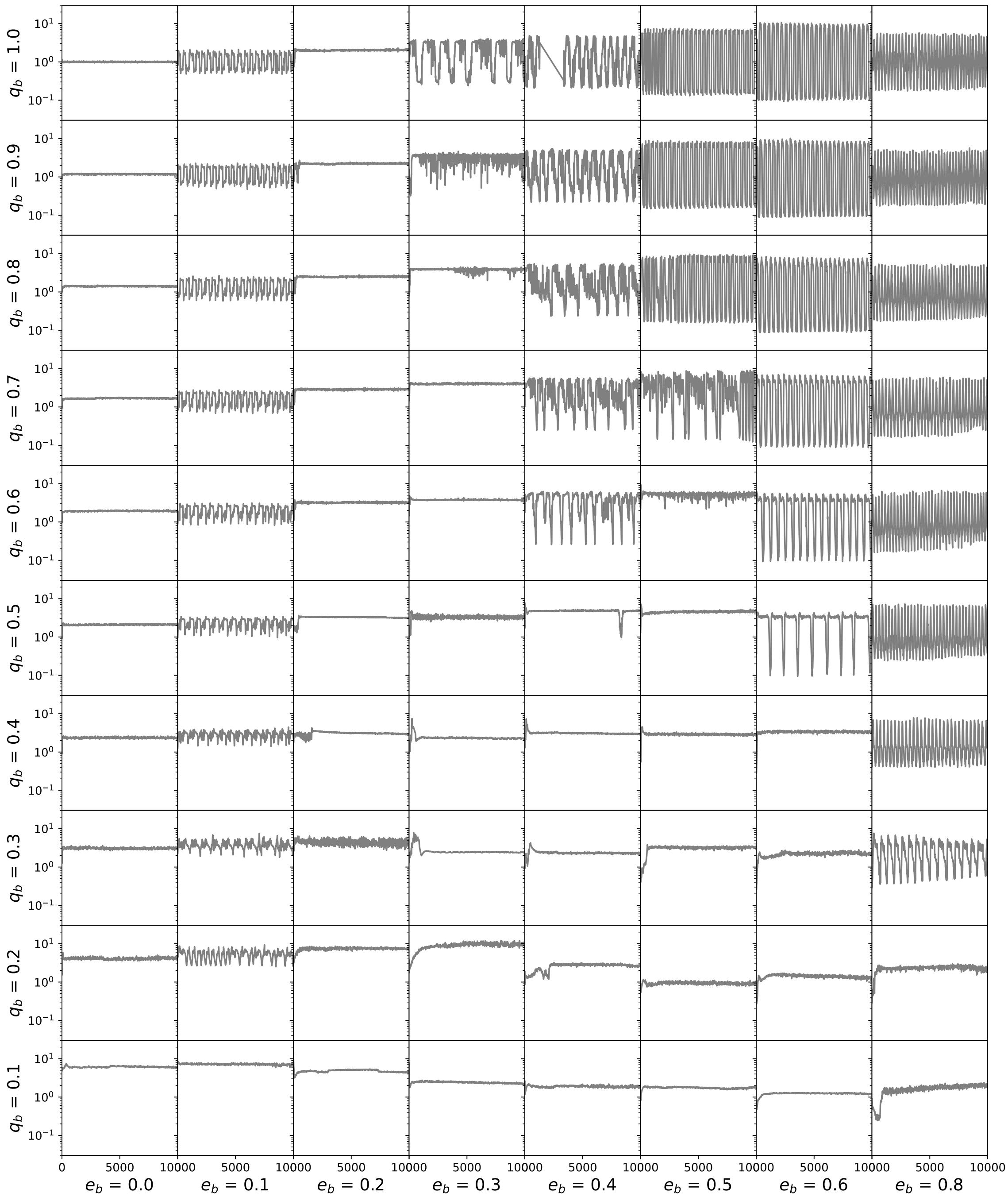

Figure 1. The accretion-rate ratio $\lambda(t) \equiv \dot{M}_2(t)/\dot{M}_1(t)$ for the entire 80-simulation suite. Each panel is a unique (e_b, q_b) simulation, with $\lambda(t)$ on the y-axis and time on the x-axis (in units of binary orbital periods τ_b). The panel's position on the larger grid signifies its binary parameters: e_b is constant along columns and increases left-to-right; q_b is constant along rows and increases bottom-to-top.

Figure 2. Heat-map of σ_λ, the standard deviation of $\lambda(t)$ over the post-instability ($t > 3000\tau_b$) portion of the simulation. Larger values (brighter) indicate stronger time-variability of preferential accretion. The variability peaks broadly in the high-e_b, intermediate-q_b region of parameter space.

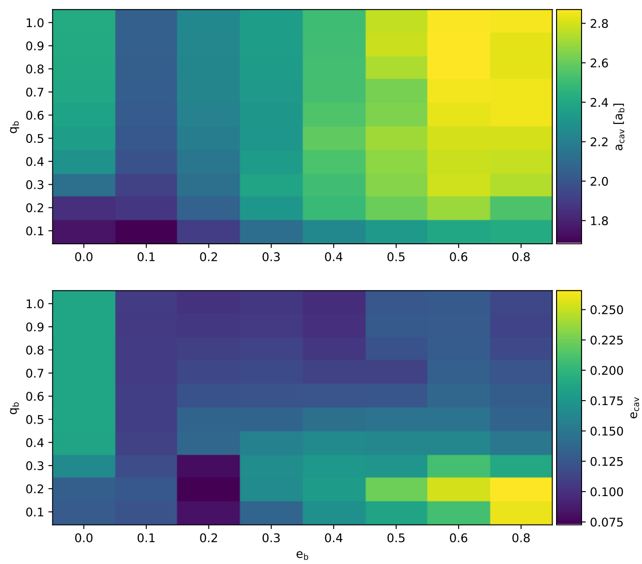

Figure 3. Reproduction of Figure 18 from DeLaurentiis & Rafikov (in preparation), displaying the semi-major axis and eccentricity of the cavities of our simulations with varying binary eccentricity and mass ratio. The cavity is characterized by the $m = 1$ Fourier mode of the CBD cavity

while eccentric, forced-precessing binaries should display periodically fluctuating preferential accretion. Our Table 1 extends this picture systematically across the full (e_b, q_b) grid, mapping every simulation in the suite onto the time-stable / time-varying dichotomy and matching it directly to the locked / precessing partition of the CBD.

The $e_b = 0$ simulations are the natural test of this picture: their CBDs precess freely yet their $\lambda(t)$ is constant. This is consistent with S23's symmetry argument — the azimuthal symmetry of a circular binary orbit prevents the precessing CBD from imprinting its variability on the relative accretion rates, even though the disk itself is precessing. It is also consistent with DeLaurentiis et al. 2025, who studied a different but analogous setup: instead of varying e_b at fixed (non-precessing) binary, they fixed the binary on an eccentric orbit and imposed a general-relativistic apsidal precession on the binary itself. They found that the GR precession of the binary's pericenter introduces a dominant modulation in the accretion rate, but only when the binary is eccentric enough that the pericenter direction matters; circular binaries cannot register the precession at all. Our finding here — that $e_b = 0$ binaries fail to develop $\lambda(t)$ variability despite a freely precessing CBD — is the disk-precession analogue of their binary-precession result: in both cases, a non-zero binary eccentricity is required for any precession (of the disk or of the binary) to imprint itself on the relative accretion rates.

Further, we note that the amplitude of the $\lambda(t)$ oscillations are not uniform among the time-varying simulations. In fact, we note that there is a clear correlation between the amplitude of the $\lambda(t)$ oscillation and the e_b of the binary.

In Fig. 2 we display a heat-map indicating the variability amplitude of $\lambda(t)$, the standard deviation σ_λ over the post-transient window $3000 \le t/\tau_b \le 10\,000$ ($N = 700$ samples at the $10\tau_b$ snapshot cadence)[3]. We note that while many of our simulations have small σ_λ since they are time-stable (see Table 1), those that display meaningful σ_λ suggest a trend. Namely, we find that σ_λ increases with e_b, peaking

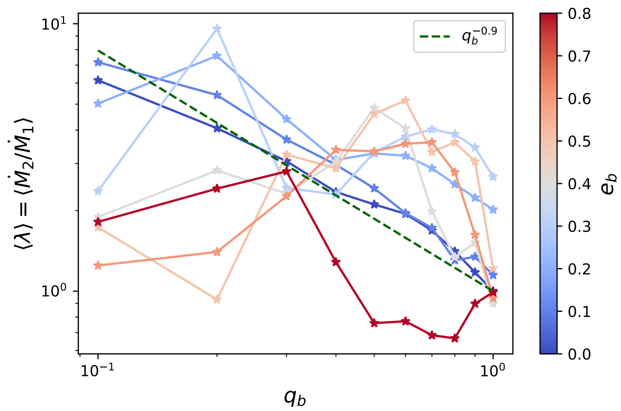

Figure 4. Time-averaged accretion-rate ratio $\langle\lambda\rangle = \langle\dot{M}_2/\dot{M}_1\rangle$ as a function of q_b (x-axis) and e_b (line color). The gray-shaded region marks $\langle\lambda\rangle > 1$, where the simulation labelled "secondary" accretes preferentially. The dashed green line is the $q_b^{-0.9}$ power-law fit reported by S23 for circular binaries.

at $e_b = 0.6$. This modest increase in σ_λ with e_b is reminiscent of Fig. 3, Figure 18 of DeLaurentiis & Rafikov (in preparation), which indicates that the eccentricity of the cavity increases with e_b and peaks at $e_b = 0.6$. This correlation further suggests that the CBD is a crucial regulator of preferential accretion.

In addition to temporal nature of $\lambda(t)$ we also comment on the mean value of $\lambda(t)$, determining which BH is preferred to accrete and to what extent. We report our results in Fig. 4 and Fig. 5.

In Fig. 4 we display the mean values of $\lambda(t)$ on the y-axis and the q_b on the x-axis, while the color represents the e_b. We find that our figure is in broad agreement with Figure 4 of S23. In particular, $\langle\lambda\rangle$ of the $e_b = 0$ simulation seems to follow the power law of $q_b^{-0.9}$. However, we also find that $e_b = 0.1$ also holds true to this line, suggesting that the behavior of $\langle\lambda\rangle$ is generic to lower eccentricity

[3] The first $3000\tau_b$ of each time-series are discarded to remove the initial disk-instability transient documented in DeLaurentiis & Rafikov (in preparation); the same cut is applied to every $\lambda(t)$ statistic reported in this paper.

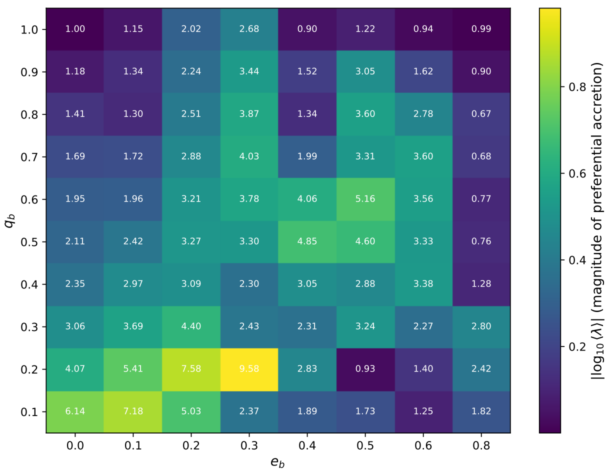

Figure 5. Heat-map showing the magnitude of preferential accretion across the suite. The colormap encodes $|\log_{10}\langle\lambda\rangle|$, the deviation of the time-averaged accretion-rate ratio $\langle\lambda\rangle = \langle\dot{M}_2/\dot{M}_1\rangle$ from unity (equal-rate accretion); brighter cells indicate stronger preferential accretion regardless of which BH happens to accrete more. In-cell labels show the actual $\langle\lambda\rangle$ value: $\langle\lambda\rangle > 1$ in 71 of 80 cells (the secondary out-accretes the primary, the standard preferential-accretion picture); $\langle\lambda\rangle < 1$ in the remaining 9 cells, of which most are $q_b = 1$ simulations where the apocenter labelling convention can be assigned either way (so $\langle\lambda\rangle$ slightly above or below unity is equally probable), plus a small number of unequal-mass dimspots (notably $(e_b, q_b) = (0.5, 0.2)$ at $\langle\lambda\rangle = 0.93$) where the asymmetry is anomalous and discussed below.

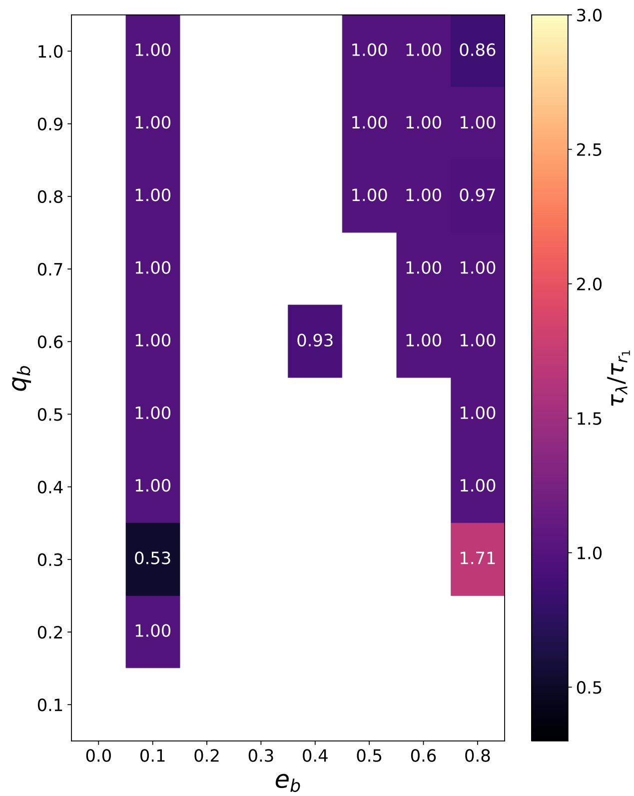

Figure 6. The ratio of the period of preferential accretion variability (τ_λ) to the period of the CBD precession, τ_{r_1}. The x-axis is the eccentricity of the binary, the y-axis is the mass-ratio of the binary, and the color of the cells correspond to the ratio of the periods. The uncolored cells are those where a clear period for r_1 or λ could not be determined.

binaries and not unique to $e_b = 0$. Further, all of our unequal-mass simulations suggest that if there is preferential accretion it is always to the secondary BH. This is in line with all the results of the literature (Muñoz et al. 2019; Duffell et al. 2020; Farris et al. 2014; Dittmann & Ryan 2021; Siwek et al. 2023a). Despite preferential accretion being biased towards the secondary, we note that the extent to which one BH accretes over the other depends greatly on the binary parameters.

The behavior of $\langle\lambda\rangle$ is perhaps best shown in Fig. 5, which displays q_b on the y-axis and e_b on the x-axis, encoding the magnitude of preferential accretion by color. At first glance it is clear that some binary parameters lend themselves to stronger preferential accretion than others: low-e_b, low-q_b binaries display the largest values, while high-e_b, high-q_b binaries are weaker, in agreement with S23. This is the opposite of the trend we saw for σ_λ values in Fig. 2 and for the eccentricity of the CBD (DeLaurentiis & Rafikov, in preparation). This suggests that the binary parameters themselves are more important than the CBD in regulating the level of preferential accretion.

While there is certainly a trend, the level of preferential accretion is by no means monotonic with respect to q_b and e_b. We note that intermediate values (e.g. $(e_b, q_b) = (0.4, 0.4)$ and $(0.5, 0.5)$) have higher preferential accretion levels than their neighbors. Further, Fig. 5 displays interesting "hotspots" of high preferential accretion, with $(e_b, q_b) = (0.3, 0.2)$ being the simulation with the highest levels of preferential accretion. We also find curious dimspots where the accretion onto both BHs is near-equal despite the binary having unequal mass. Notably, $(e_b, q_b) = (0.5, 0.2)$ has $\langle\lambda\rangle = 0.93$.

More importantly, we would like to note a point of deviation from S23. In Fig. 4 and Fig. 5 we report $\langle\lambda\rangle \neq 1$ for the $(e_b, q_b) = (0.2, 1.0)$ and $(0.3, 1.0)$ simulations. The $(e_b, q_b) = (0.2, 1.0)$ simulation in S23 had a numerical error, resulting in a deviation in λ. Accounting for this, we determine that for the $(e_b, q_b) = (0.2, 1.0)$ and $(0.3, 1.0)$ simulations, the binary, in fact, preferentially accretes onto one of the BHs. The implications of this are further discussed in Section 3.2 and Section 4.

Clearly, there is a variability in both the time behavior of preferential accretion and its value across simulations. In the following section we make a preliminary attempt to understand preferential accretion as an effect of the unique geometry between the binary and the CBD.

3.1.1 *Preliminary analysis*

In Section 3.1 we noted similarities between the preferential accretion behavior $\lambda(t)$ and the CBD geometry. The delineation of time-stable and time-varying $\lambda(t)$ in Table 1 is similar to that between precessing and locked circumbinary disks. σ_λ seemed to display the same trend with e_b and q_b as that of the CBD eccentricity. While alternative explanations have been proposed (Artymowicz 1983), the assumption that the preferential accretion of the binary is related to the relative closeness of the components to the CBD's inner edge (Rafikov 2016; Farris et al. 2014) has been left unchallenged. In the following section we provide a first, preliminary, analysis of this hypothesis and alternative ways in which the CBD would influence preferential accretion.

As discussed in Section 2, we use the dominant Fourier modes of the cavity to reconstruct its shape as a function of azimuthal angle θ: $r_{\rm cav}(\theta)$, with the origin at the binary center of mass. Combined

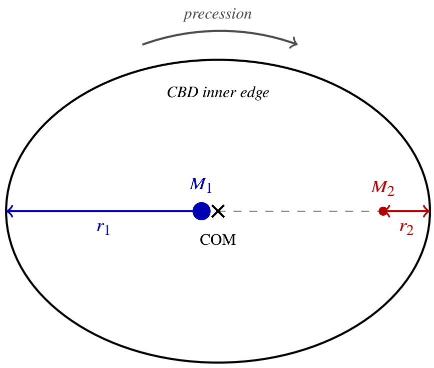

Figure 7. Schematic of the cavity geometry at apocenter for the locked-CBD case, illustrated for an unequal-mass binary with $q_b \ll 1$. The CBD inner edge (black ellipse) is closer to the secondary BH (M_2, red) on one side; the primary (M_1, blue) is farther from its nearby cavity wall. Closest distances from each BH to the wall are r_1 and r_2, with $r_1 > r_2$ in the locked configuration shown. As the disk precesses (curved gray arrow above the ellipse), the ellipse's major axis rotates, and the orientations of r_1 and r_2 change accordingly.

with the position of each BH at the apocenter snapshot, we define the shortest distance from the cavity inner edge to each BH. We take θ_1^* and θ_2^* to be the angles that minimize the distance from each BH to the cavity wall, and we then write $r_1 = r_{\rm cav}(\theta_1^*) - r_{\rm BH_1}$ and $r_2 = r_{\rm cav}(\theta_2^*) - r_{\rm BH_2}$ for primary and secondary respectively (see Fig. 7). We emphasise that our snapshots are taken only at apocenter; r_1 and r_2 are not literal closest-approach distances over the full orbit but a single-phase snapshot of the cavity orientation relative to the binary axis. As the disk precesses, the orientation of $r_{\rm cav}(\theta)$ relative to the fixed apocenter line changes, and $r_1(t)$ and $r_2(t)$ inherit that variability. They therefore serve as a proxy for cavity orientation, not as a moment-by-moment proximity metric.

A first step in determining the relationship between preferential accretion and the CBD is to study the temporal behavior of these quantities. We have already determined that, except for the $e_b = 0$ case, all precessing CBDs result in a time-varying $\lambda(t)$. $r_1(t)$ and $r_2(t)$, proxies for the CBD orientation, display the same split into time-stable and time-varying behavior as $\lambda(t)$ in Table 1. For simulations with time-varying $\lambda(t)$ we compare their period of oscillation with that of r_1 and r_2. To determine the period of each time-series we first calculate a fast Fourier transform (FFT) and normalize the amplitudes by their sum. The period is the defined at that of the largest amplitude in the spectrum which has a normalized value > 0.05. We find the periods of r_1[4] and λ in this fashion and display the ratio of the two periods in Fig. 6.

The ratios reported in Fig. 6 are nearly all equal to unity. Though certain binaries have ratios that deviate from unity, these deviations do not suggest resonances between CBD precession and preferential accretion but rather point to the FFT being unstable when applied to periodic, non-sinusoidal $\lambda(t)$ (e.g. $(e_b, q_b) = (0.4, 0.8)$). Thus, the precession of the CBD and the variability of $\lambda(t)$ are tightly correlated in both occurrence and period: we observe the same FFT period in both quantities and the same locked-vs-precessing partition.

We pause to address a methodological subtlety. Our snapshots are recorded only at apocenter, every 10 binary orbital periods. This cadence is fast enough to resolve the cavity precession across our suite, since the precession period is always much longer than 20 orbits. Each snapshot therefore catches the cavity at a slightly rotated orientation, and the time-series $r_1(t)$ and $r_2(t)$ inherit the cavity's precession frequency. Two empirical checks confirm this. First, in every simulation r_1 and r_2 share the same FFT period — which is exactly what we expect if both are sampling the same precessing cavity from different sides. Second, that shared period matches the period of $\lambda(t)$ (Fig. 6). Fig. 8 makes both checks visible: for $(e_b, q_b) = (0.8, 0.3)$ the three signals each peak cleanly at $\tau \approx 700\tau_b$ above our 0.05 amplitude threshold, while $(0.4, 0.7)$ shows the same period concentration but at lower amplitudes — the regime in which the FFT becomes unstable when applied to non-sinusoidal $\lambda(t)$. The period match between $\lambda(t)$ and $r_1(t)$, $r_2(t)$ therefore licenses the inference that $\lambda(t)$ variability is paced by the cavity's apsidal precession, even though the precession itself is not directly resolved at our snapshot cadence.

In Fig. 9 we display r_1 (blue) r_2 (red) and λ (black) for a slice in time for six of our simulations, with the dual y-axis representing the dimensionless lambda value (left) and the distance from each BH to the edge of the cavity in units of a (right). As indicated by Fig. 6 the variability of the distance from the BHs to the cavity displays the same period (or lack thereof) as λ. This is clear for the simulations in the left column of Fig. 9. However, we note that for some of the simulations in the right column, where the period is unstable. This behavior is especially well captured by $(e_b, q_b) = (0.5, 0.7)$, where the change from a stochastic to periodic λ signal at $\approx 8500\tau_b$ is mirrored in the change of behavior of r_1 and r_2. Further, in $(e_b, q_b) = (0.4, 0.7)$ we find that large amplitude changes in r_1 and r_2 (e.g. at $t \approx 6000$ and $9000\tau_b$) occur simultaneously with large amplitude changes in λ.

The naive understanding of the tie between preferential accretion and the distance to the black holes is as follows. The inner cavity rim hosts the densest, slowest-moving gas in the system (Shi et al. 2012; Farris et al. 2014; Duffell et al. 2020). Whenever a black hole's Hill sphere overlaps that rim, the steep local gravitational-potential gradient bends incoming streamlines into its mini-disk, and the instantaneous capture rate obeys a Bondi-Hoyle-like scaling

$$\dot{M}_{\rm near} \propto \Sigma_{\rm rim}\, v_{\rm rel}^{-3}. \tag{4}$$

A companion farther from the rim encounters lower surface density and higher gas-BH relative velocity (Miranda et al. 2017; D'Orazio & Duffell 2021), so

$$\frac{\dot{M}_{\rm far}}{\dot{M}_{\rm near}} \sim \frac{\Sigma}{\Sigma_{\rm rim}} \left(\frac{v_{\rm rel}}{v_{\rm rel,\,near}}\right)^3 \ll 1, \tag{5}$$

suppressing its accretion by orders of magnitude even though both black holes share the same global gas reservoir.

In $(e_b, q_b) = (0.2, 0.1)$ we see what we would expect from our naive explanation. The corresponding panel of Fig. 9 shows that this simulation has a CBD that is locked such that our primary black hole (blue) is about three times as far from the cavity edge than our secondary black hole (red). Correspondingly, we see that the secondary is accreting at a rate about five times the primary, seemingly as a result of the secondary's greater pull on its neighboring portion of the CBD. In fact, we can even take this further and note that the tidal field of the secondary on its nearby cavity wall is approximately four to five times stronger than the tidal field of the primary on its more distant cavity wall (with the tidal field scaling as M/r^3 and $r_1/r_2 \approx 3.4$ measured from the simulation snapshot), in agreement with the observed factor of ~ 5 in the relative accretion rate. However, our naive approximation is inherently over-simplistic. It doesn't take into account each BH's minidisk, the streams by which the gas is fed to the BH, or any non-linear fluid dynamics, such as shocks,

[4] It is important to note that we have found the period of r_1 and r_2 to be exactly equal in all our simulations, making either appropriate for this calculation.

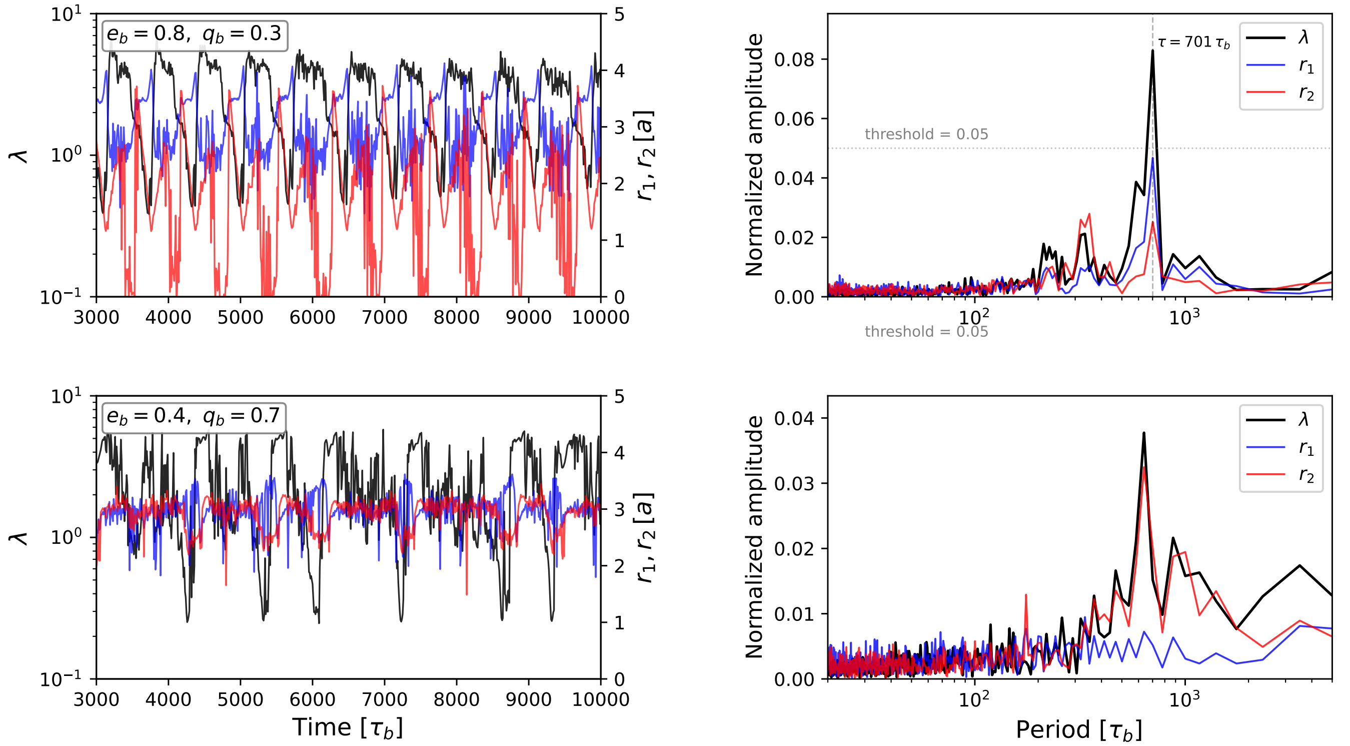

Figure 8. Time-series and power spectra for the two messy cases of Fig. 9: $(e_b, q_b) = (0.8, 0.3)$ (top row) and $(0.4, 0.7)$ (bottom row). Left column: the same $\lambda(t)$ (black, left log-axis), $r_1(t)$ (blue), and $r_2(t)$ (red, right axis in units of a), where r_i is the shortest distance from BH i to the cavity inner edge at the apocenter snapshot, shown over the post-transient window $3000 \le t/\tau_b \le 10000$ ($N = 701$ samples at the snapshot cadence of $10\tau_b$). Right column: normalized rFFT amplitudes $|\tilde{X}(f)|/\sum_f |\tilde{X}(f)|$ as a function of period $\tau = 1/f$; the dotted line marks the 0.05 amplitude threshold used in Fig. 6. For $(0.8, 0.3)$ all three signals peak at $\tau \approx 700\tau_b$, with λ's peak well above threshold; for $(0.4, 0.7)$ the peak sits in the same period range but below threshold, illustrating why the period ratio in Fig. 6 departs from unity for non-sinusoidal $\lambda(t)$.

that may be relevant in this mechanism. Studying the other panels, we see that our naive explanation no longer suffices.

Before discussing the deviations, we make the naive prediction explicit. If preferential accretion were governed solely by proximity to the cavity wall, then for a binary at apocenter with the cavity oriented toward the secondary, r_2 should be at its minimum (cavity wall closest to secondary) precisely when λ is at its maximum (secondary accreting most). As the cavity precesses, r_2 should rise to its maximum a half-precession-period later, when λ should be at its minimum. We therefore expect $r_2(t)$ to be *exactly π out of phase* with $\lambda(t)$, while $r_1(t)$ — by symmetry, since the cavity wall is then closest to the primary — should be *exactly in phase* with $\lambda(t)$.

All simulations displayed in Fig. 9, except for $(e_b, q_b) = (0.2, 0.1)$, show that lower r_2 values do not correspond to higher accretion rates onto the secondary. The simulations $(e_b, q_b) = (0.5, 0.8)$ and $(0.8, 1.0)$ clearly do not display the naive behavior: the former shows both r_1 and r_2 at $\pi/4$ out of phase with λ, while the latter shows the exact opposite of our naive expectation, with r_1 in phase and r_2 out of phase with λ. Further, $(e_b, q_b) = (0.8, 0.3)$ and $(0.4, 0.7)$ break the naive picture in another way: the amplitude of r_1 relative to r_2 does not strictly coincide with which BH accretes preferentially. For $(0.8, 0.3)$, while the primary is always at least as far from the cavity wall as the secondary, we still see periods where the primary accretes preferentially. For $(0.4, 0.7)$, during periods where both BHs are equidistant from the cavity wall, the secondary accretes up to 4 times more than the primary[5].

From the above analysis it is clear that the naive explanation of distance to the cavity and tidal field is not robust. The variability and periods of $\lambda(t)$ and r_1, r_2 are correlated, but with no clear causal link between them. While this study has focused on describing the plethora of preferential accretion behaviors and their ties to the CBD, a more robust explanation of how the binary preferentially accretes will be explored in a forthcoming paper.

3.2 Mass ratio

In addition to preferential accretion, we also report the average rate of change of the mass ratio, $\langle \dot{q} \rangle$. Namely, we calculate $\dot{q}(t)$ as described in equation 1 in Section 2, make a time-cut at 3000τ to ensure that early instabilities do not effect our results, and report the mean on the truncated time-series in units of $\dot{M}_b/M_b$ in Fig. 10.

A striking feature of Fig. 10 is that for all $q_b \neq 1$ simulations the binary $\langle \dot{q} \rangle$ is positive, meaning that the binary is evolving towards equal mass. We note, however, that the rate of this evolution is quite varied, spanning two orders of magnitude. Further, the gradient of $\langle \dot{q} \rangle$ in parameter space is not uniform. In fact, we see that there exists

[5] While λ is a ratio and this behavior could be due to small amplitude deviations in $\dot{M}_1$ and $\dot{M}_2$, we found that the amplitude of the sum is not strongly suppressed during this period.

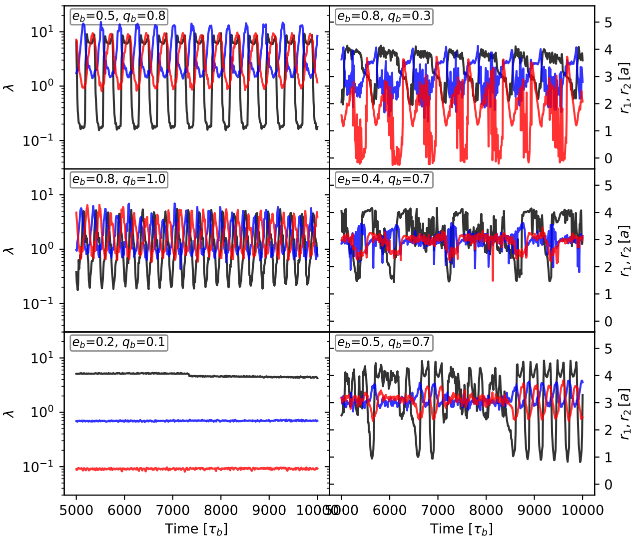

Figure 9. Comparison between the time-series of $\lambda(t)$ (black, left log-axis), $r_1(t)$ (blue), and $r_2(t)$ (red) (right axis, in units of the binary semi-major axis a), where r_1 and r_2 are the shortest distances from the primary and secondary, respectively, to the cavity inner edge at the apocenter snapshot (a proxy for cavity orientation; see §3.1.1). Time is in binary orbital periods τ_b; panels are plotted for $5000 \le t/\tau_b \le 10000$. Clockwise from upper-left: $(e_b, q_b) = (0.5, 0.8)$, $(0.8, 0.3)$, $(0.4, 0.7)$, $(0.5, 0.7)$, $(0.2, 0.1)$, and $(0.8, 1.0)$. We find that the BHs' position with respect to the cavity does not determine the components' relative accretion rates.

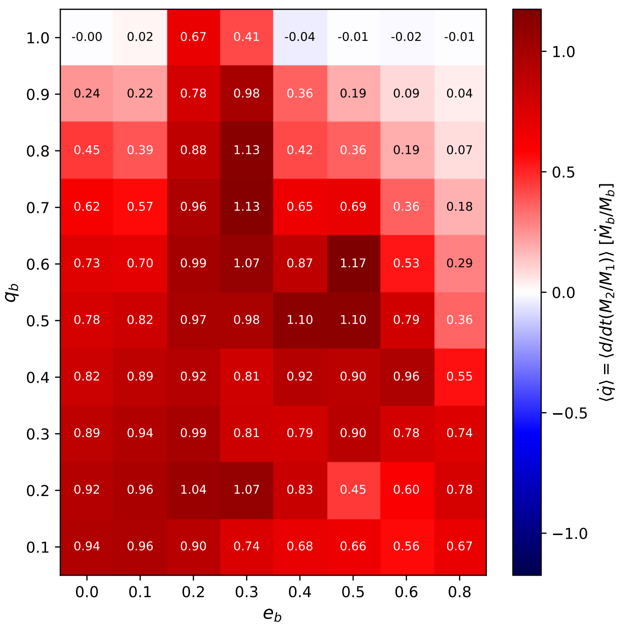

Figure 10. The time-averaged rate of change of the binary mass-ratio, $\langle\dot{q}\rangle$, for the simulation suite, in $\dot{M}_b/M_b$ units. The x-axis is the binary eccentricity e_b and the y-axis is the binary mass-ratio q_b. The diverging colormap is centered at $\langle\dot{q}\rangle = 0$: red cells indicate the binary evolves toward $q = 1$ (positive $\langle\dot{q}\rangle$), blue cells away from $q = 1$. Most $q_b < 1$ cells are positive (driven toward equal mass); the $q_b = 1$ row is mostly consistent with zero, with notable exceptions at $e_b = 0.2$ and 0.3 where the binary evolves away from unity (§3.2).

a massive drop off in $\langle\dot{q}\rangle$ in the upper right corner of Fig. 10, starting along the diagonal extending from $q_b = 0.8, e_b = 0.6$.

Turning to the upper row of Fig. 10, the result is best framed as the presence or absence of mass-ratio evolution among initially equal-mass binaries: most $q_b = 1$ simulations display $\langle\dot{q}\rangle$ values consistent with zero[6] (absence of evolution), while $(e_b, q_b) = (0.2, 1.0)$ and $(0.3, 1.0)$ both display strong positive $\langle\dot{q}\rangle$ (presence of evolution away from unity).

We pause to clarify a notation subtlety. Our convention $q_b \equiv M_2/M_1 \le 1$ assigns "primary" to the more massive BH and "secondary" to the less massive one. In a strictly $q_b = 1$ system this assignment is degenerate, and we adopt the convention — following S23 — of identifying the components by their spatial location at apocenter. Once $q_b = 1$ is broken by accretion, the BH initially labeled "secondary" grows into the more massive component, formally inverting the $q_b \le 1$ convention. We continue to label the BHs by their initial assignment throughout the integration for clarity. The statement that the binary "evolves away from unity" should be read as: the mass ratio M_2/M_1 deviates from unity, where the labels 1 and 2 refer to the original assignment, even after the inversion.

This result for $(e_b, q_b) = (0.2, 1.0)$ stands as a correction to S23, where it was reported that q_b would remain at unity for this case. After ensuring correct sink-particle tracking in the $q_b = 1$ simulations, we find that accretion toward unity for SMBBHs is not necessarily a given. This makes sense in light of the results of DeLaurentiis & Rafikov (in preparation), where it was reported that the $(e_b, q_b) = (0.2, 1.0)$ and $(0.3, 1.0)$ simulations have stable, locked disks in line with S23, with a stable non-varying λ, which in this case was coincident with the primary being further from the cavity edge than the secondary.

The finding that equal-mass binaries at $e_b = 0.2$, 0.3 accrete away from equal mass has implications for CBD structure and SMBBH population statistics. From DeLaurentiis & Rafikov (in preparation) we note that both $(e_b, q_b) = (0.2, 1.0)$ and $(0.2, 0.9)$ simulations have locked disks in roughly the same orientation, with the pericenter of the disk closest to the secondary. As the $q_b = 1$ case accretes away from unity, the BH initially identified as the "secondary" grows into the primary; the disk, oriented toward the original secondary, must therefore realign itself, flipping in concert with the switch in primary and secondary identities. We speculate that this realignment proceeds on the disk's apsidal precession timescale, since the same precession dynamics that orient locked disks in the first place are the natural mechanism by which a locked disk can re-orient. The details of this reaction would provide insight into the CBD-orientation mechanism, into how far from unity the binary ultimately evolves, and — depending on the geometry and timescale of re-orientation — could constitute an event with characteristic EM signatures. Confirming this picture would require live-binary simulations through a sustained $\dot{q} \neq 0$ phase, which we leave to future work.

4 OBSERVATIONAL IMPLICATIONS

In the following section we discuss the observational consequences of our $\lambda(t)$ and $\dot{q}$ results (see Section 3).

[6] The slightly negative values are within the standard error of the mean for the $\dot{q}$ time-series ($\sigma \approx 10^{-2}$ for $q_b = 1$, estimated from the variance of $\dot{q}(t)$ divided by the integration duration), and can thus be taken to be zero.

4.1 Flickering jets

A key observational consequence of accretion onto BHs is the potential launching of jets. While the exact mechanism continues to be an extremely active area of study, jet launching can be broadly understood through the radiative efficiency of the accretion disk. When the disk is radiatively efficient, photons (and the energy they carry) are efficiently radiated away. In radiatively inefficient disks, photons are trapped, building up thermal and magnetic pressure. This scenario can produce large pressure gradients capable of driving jets, and a strong magnetic field configuration whose energy can be extracted via the Blandford-Znajek mechanism. Radiatively inefficient, geometrically thick disks are commonly found at low accretion rates $\dot{M} < 0.01\dot{M}_{\rm Edd}$ (Guolo et al. 2021) and at high accretion rates $\dot{M} > \dot{M}_{\rm Edd}$, where $\dot{M}_{\rm Edd}$ is the Eddington accretion rate — the steady rate at which radiation pressure on free electrons exactly balances gravity, formally defined in Eq. 6 below. In both regimes, the geometric thickness of the disk plays a second role beyond setting the radiative efficiency: the inflated inner walls form a funnel along the BH spin axis that channels magnetic flux and outgoing material into a collimated relativistic jet. Given these criteria, we are able to make statements about the presence or absence of jets from our simulations.

To do so, we must first scale our numerical accretion rates, which are in units $M_{\rm bin}/\tau$, to Eddington units. For a BH of mass M, the Eddington accretion rate is given by

$$\dot{M}_{\rm Edd} \equiv \frac{4\pi G M m_p}{c\eta\sigma_t} \tag{6}$$

where m_p is the proton mass, σ_t is the Thomson scattering cross-section and $\eta \equiv 0.1$ is the radiative efficiency. To retain information about the relative accretion rates of the two BHs, we set the binary accretion rate $\dot{M}_b \equiv \dot{M}_1 + \dot{M}_2$ equal to $\gamma\dot{M}_{\rm Edd}$ (with γ an arbitrary scale factor evaluated at the total binary mass), and convert the accretion rate of each individual BH to its own Eddington units.

In Fig. 11 we set the accretion rate of the binary to be $1\dot{M}_{\rm Edd}$ assuming a $10^7 M_\odot$ binary and plot the accretion rate for both BHs normalized to their respective Eddington accretion rates. The horizontal gray dashed line is at $1\dot{M}_{\rm Edd}$ to represent the accretion rate above which which jets are likely to launch. The red lines represent the accretion rate of the secondary, and the black lines represent the accretion rates of the primary. The background color of the panel is associated with different jet-behaviors: purple for dual jets, blue for a single jet, green for flickering jets. The time-slice displayed is arbitrary and serves to merely highlight the accretion behavior.

A striking feature of Fig. 11 is the wide variety in magnitude between the two BHs' accretion rates. Since the Eddington accretion rate scales linearly with the BH mass we expect that the accretion rates of the secondary to be increased greatly when normalized to Eddington units. This is evidenced by the nearly 2 order of magnitude difference between the secondary and primary at $q_b = 0.1$. Further, we also notice that the behavior of the individual accretion rates of the black holes are quite varied, as the $\lambda(t)$ results suggested. Aside from the differences in whether the BHs' accretion rates are stable or not, the profile of the accretion rate itself is varied. Some binaries experience accretion rates that are close to sinusoidal (e.g $e_b = 0.6, 0.8$), others seem to closer resemble square-waves (e.g. $e_b = 0.3$ and 0.5 at $q_b = 1$), others yet have quite sharp breaks that evade simple characterizations (e.g. $(e_b, q_b) = (0.1, 0.7)$). Further, we note that the accretion rate of one BH is not always simply the accretion rate of the other with a different baseline and $\pi/2$ phase shift. Rather, they can take on notably different profiles from each other, resulting, at times, in both BHs experiencing a local peak in accretion rate, but because of different accretion rate amplitudes result in a peak in λ. It is this plethora of individual BH accretion rates, and the way in which they compare to each other, that yield an interesting assortment of jet-launching behaviors.

In Fig. 11 we delineate three broad regimes: a) binaries where one BH launches a jet (blue), b) binaries where both BHs coincidentally launch jets in a sustained and repeated fashion (purple), c) binaries where both BHs launch jets in a successive, alternating fashion (green). We describe these jet-behavior regimes as single jets, dual jets, and flickering jets, respectively. We assigned jet-regimes by determining whether the accretion rate of each BH surpassed a threshold value of $1.1\dot{M}_{\rm Edd}$ for greater than 50τ — a threshold chosen modestly above unity to allow the disk thickness to inflate enough to support the funnel collimation discussed above — and whether those instances were temporally coincident for greater than 50τ. The 50τ duration was chosen empirically: it is long enough to filter out short-lived threshold excursions (single-orbit transients, accretion bursts) and require a sustained launching episode, but short enough to preserve the alternating cadence we want to detect in the flickering regime. Jet activity is itself expected to follow the inner-disk dynamical time $t_{\rm dyn} \sim \Omega_K^{-1}$, which for our 2D setup is of order a binary orbital period; the 50τ window therefore samples many dynamical times.

We note that the majority of simulations at the assumed binary accretion rate of $1\dot{M}_{\rm Edd}$ are single-jet systems, where most of them have low q_b due to the highly unequal Eddington-normalized accretion rates of those systems.

The flickering jet systems are clustered at higher q_b and e_b. Due to large λ amplitudes, such systems are able to exist up to $q_b = 0.5$. It is important to note that since these flickering jets are dependent on large λ oscillations they are unique to $e_b \neq 0$ binaries.

Finally, we have a few dual-jet systems. For those at $e_b = 0$ both BHs accrete at nearly the same, stable rate and thus both launch sustained jets. However, we also note that we have dual jets for binaries whose $\lambda \neq 1$. We note that the dual-jet systems at $e_b = 0.1$ and $e_b = 0.8$ show both BHs having coincident, local peaks in accretion rates, meaning that the binary sustainably and periodically launches two jets.

While dual jets from BBH systems have been suggested before (Palenzuela et al. 2010; Qian et al. 2019) and demonstrated numerically in a range of GR-MHD setups (Gutiérrez et al. 2024; Ressler et al. 2025; Ruiz et al. 2023; Most & Wang 2024; Ennoggi et al. 2025), we believe that we are *the first to report evidence suggesting a flickering-jet behavior* — alternating, rather than coincident, jet activity from the two BHs. We emphasize that the prior GR-MHD work has typically begun from configurations in which jet conditions are already satisfied at both BHs (e.g., a magnetically arrested circumbinary flow) and asks how the resulting jets evolve through inspiral. Our framing is complementary: we ask, given a hydrodynamic accretion-rate time series, when each BH's accretion crosses the threshold required for jet activity in the first place. The flickering regime is unique to $e_b \neq 0$ binaries with large λ amplitudes, and provides a uniquely distinct feedback mechanism that could be further explored in cosmological simulations.

Fig. 11 and its discussion assumed a binary accretion rate of $1\dot{M}_{\rm Edd}$. However, this assumption only controls the baseline of the binary accretion rate. While a different assumed binary accretion rate would not change the profiles nor the relative accretion rates of the components, it would affect the absolute amplitude — and, given that jets launch at *both* super-Eddington rates ($\dot{M} > \dot{M}_{\rm Edd}$, geometrically thick disk) and sub-Eddington ADAF rates ($\dot{M} < 0.01\dot{M}_{\rm Edd}$), the choice of $\dot{M}_b$ determines which regime each BH inhabits. To make

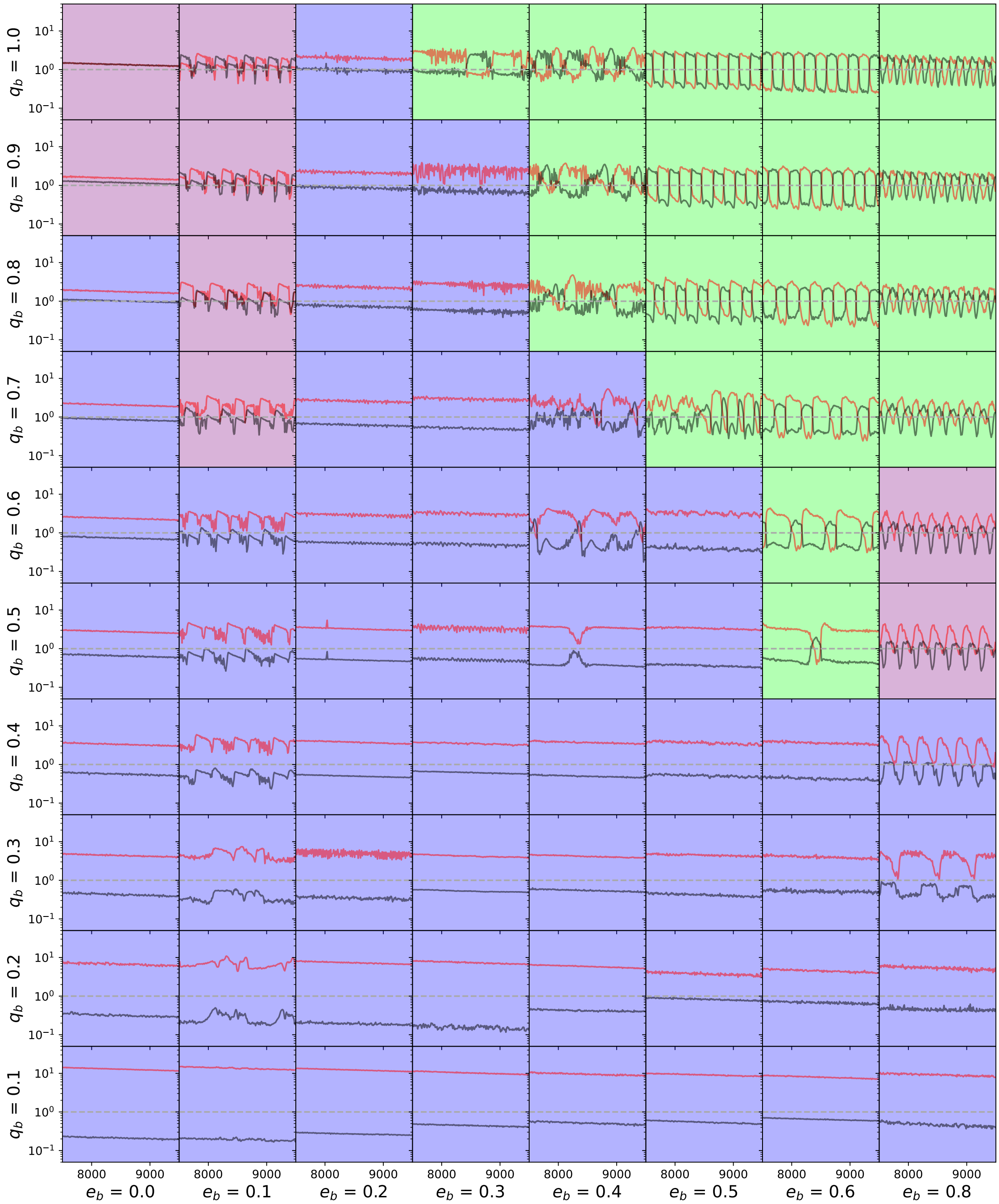

Figure 11. Depiction of Eddington-normalized accretion rates for our simulation suite, with panel backgrounds colored by jet-regime. The panel structure is the same as that of Fig. 1, with time on the x-axis of each panel and per-BH accretion rate (in Eddington units) on the y-axis. We display the accretion rate of the primary (black), the secondary (red), and the jet-launching threshold of $1\dot{M}_{\rm Edd}$ (horizontal dashed gray). Blue background colors indicate single jet regimes, purple indicates dual-jet regimes, and green indicates flickering jet regimes.

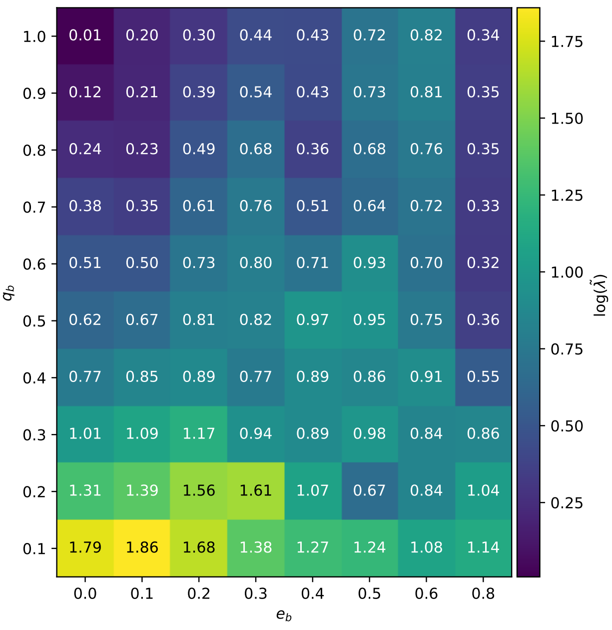

Figure 12. The Eddington-normalized accretion-rate gauge $\tilde{\lambda} \equiv \langle \max(\dot{M}_1, \dot{M}_2) \rangle / \langle \min(\dot{M}_1, \dot{M}_2) \rangle$ for our simulation suite, in $\dot{M}_{\rm Edd}$ units. The x-axis is the binary eccentricity e_b and the y-axis is the binary mass-ratio q_b; cell colors indicate $\log \tilde{\lambda}$. Combined with an assumed binary accretion rate $\dot{M}_b$, $\tilde{\lambda}$ converts directly into the Eddington-normalized accretion rate of each BH, allowing the jet-launching regime (single, dual, or flickering — see Fig. 11) to be predicted from (e_b, q_b) alone for a given $\dot{M}_b$.

this explicit, Fig. 13 reproduces Fig. 11 at two bracketing values that place each BH well inside one of the two jet-launching regimes: a deep-ADAF case ($\dot{M}_b = 0.001\,\dot{M}_{\rm Edd}$, left) and a strongly super-Eddington case ($\dot{M}_b = 5\,\dot{M}_{\rm Edd}$, right). At both extremes, dual jets dominate the parameter space, with single-jet cells surviving only at low q_b where one BH straddles the radiatively-efficient (no-jet) zone. The fiducial $\dot{M}_b = 1\,\dot{M}_{\rm Edd}$ in Fig. 11 sits in the transition band where single and flickering regimes appear because individual BHs cross the no-jet zone more readily.

Fig. 12 displays $\tilde{\lambda}$, which can be written as

$$\tilde{\lambda} \equiv \frac{\langle \max(\dot{M}_1 \dot{M}_2) \rangle}{\langle \min(\dot{M}_1, \dot{M}_2) \rangle} \tag{7}$$

where max and min are element-wise functions and the accretion rates are provided in Eddington units. The resulting $\tilde{\lambda}$ is dimensionless and provides the ratio of the maximum and minimum Eddington-normalized accretion rates of the binary components. Given the time-variability of λ in Table 1 and a binary accretion rate, $\tilde{\lambda}$ can then be easily used to determine whether one or both BHs launch a jet, and whether they are temporally coincident. Namely, one can use

$$\dot{M}_1 = \frac{\dot{M}_b}{\tilde{\lambda} + 1} \tag{8}$$

to find the accretion rate of the primary based on the scaling of λ and $\dot{M}_b$. With Fig. 12 in hand we are able to gain insight into the jet behavior of any proposed BBH system as a function of q_b and e_b. Given the clustering of jet-behavior in q_b-e_b parameter space, determination of whether a system has sustained dual jets, flickering jets, or a single jet could greatly constrain the orbital parameters of the system.

"Flickering jets" are a unique EM signature for BBHs. In order to use them to find BBH systems, we must ensure they "flicker"—i.e switch which BH is preferentially accreting—on a human time-scale. In the following we compute and place constraints on the time to "flicker."

Firstly, we require that the time to flicker τ_f be fewer than 10 years in the observer's rest-frame, so that a few cycles could be possible to find in a human lifetime. For simplicity, we assume the flickering period is 300τ. We also require that the binary not merge in less than 100 years, in order to ensure that these systems are not exceedingly rare.

In Fig. 14 we display the region of parameter space that satisfies the above time constraints (within black lines) for various binary masses at various redshifts. The x-axis of each panel is the eccentricity e_b, the y-axis is the a_b in Schwarzchild radii, and τ_f is reported in the observer frame. The gray lines represent the change in eccentricity and semi-major axis for the binary due to 10 orbits worth of GW radiation, computed via Peters 1964.

Fig. 14 shows that $10^5 M_\odot$ and $10^6 M_\odot$ systems both display considerable regions of parameter-space that satisfy the constraints necessary to observe the flicker, up to $z = 10$. We find that the region which satisfies these constraints shrinks to zero by $z = 15$ and $z = 10$ for the $10^6 M_\odot$ and $10^5 M_\odot$ systems, respectively. This provides encouraging evidence for our ability to observe these flickering jet systems.

In addition to observing a flicker occur, we note that jets are extended emission sources and thereby provide us an ability to observe a past flicker. If we could determine a geometric separation in the structure of a helical jet, this could indicate that the emission is from a binary that flickered in the past.

4.2 Unequal-mass sources

In addition to the variety of jet launching behavior and their effects on galaxy feedback, the evolution of the mass ratio of these SMBBHs is also a key observational signature. We explore whether our findings from Fig. 10 suggesting that some binaries do not evolve towards equal mass in fact lead to detectable $q_b \neq 1$ BBH systems.

In addition to its mass-ratio, the eccentricity and semi-major axis of the binary also change as functions of q_b and e_b due to the CBD, we represent these as $\dot{e}_{\rm gas}(e_b, q_b)$ and $\dot{a}_{\rm gas}(e_b, q_b)$, respectively. The values for $\dot{a}_{\rm gas}(e_b, q_b)$, and $\dot{e}_{\rm gas}(e_b, q_b)$ were also calculated for this full simulation suite and are reported in Figure 2 and 3 of Siwek et al. 2023b, respectively. Since our simulation suite is discretized in e_b and q_b space, we define $\dot{e}_{\rm gas}(e_b, q_b)$, $\dot{a}_{\rm gas}(e_b, q_b)$ and $\dot{q}(e_b, q_b)$ as sets of piecewise functions with constant values that are centered at the binary parameter values and extend to the midpoint between it and its neighboring, sequential, binary parameter value. For example,

$$\dot{q} = 1.041 \begin{cases} 0.15 \leq q_b < 0.25 \\ 0.15 \leq e_b < 0.25 \end{cases} \tag{9}$$

However, in addition to CBD effects, the GWs radiated from the system also circularize and shrink the binary; we include these effects through the standard quadrupole-formula expressions (Peters 1964)

$$\dot{a}_{\rm GW} = \frac{-64 G^3 M^3 q\, f(e)}{5 c^5 a^3 (1+q)^2} \tag{10}$$

and

$$\dot{e}_{\rm GW} = -e \frac{304 G^3 M^3 q \left(1 + \frac{121}{304} e^2\right)}{15 c^5 a^4 \left(1 - e^2\right)^{5/2} (1+q)^2}. \tag{11}$$

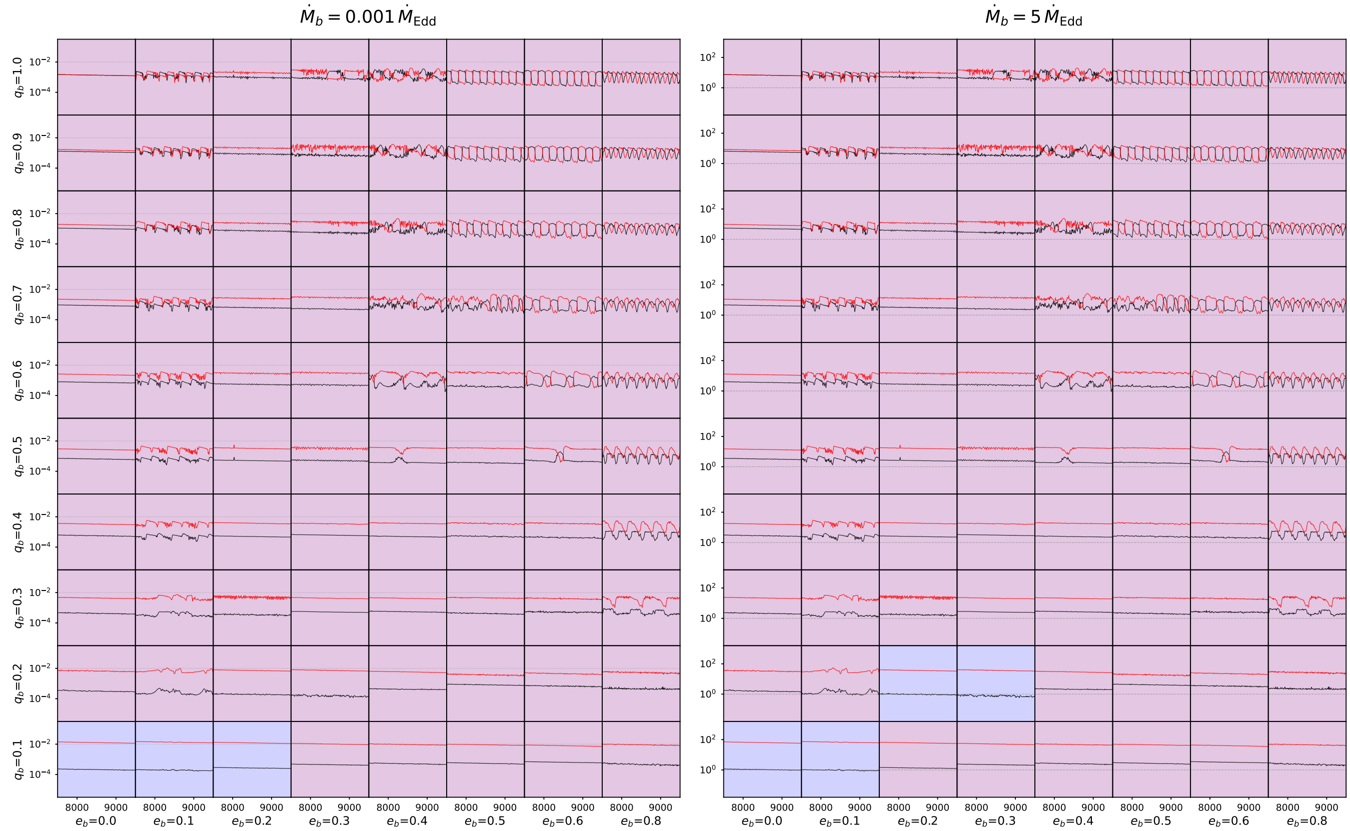

Figure 13. Reproduction of Fig. 11 at two bracketing values of the binary accretion rate that place each BH inside one of the two jet-launching regimes discussed in §4: a deep-ADAF case $\dot{M}_b = 0.001\,\dot{M}_{\rm Edd}$ (left, applying the $\dot{M} < 0.01\,\dot{M}_{\rm Edd}$ ADAF threshold) and a strongly super-Eddington case $\dot{M}_b = 5\,\dot{M}_{\rm Edd}$ (right, applying the $\dot{M} > 1.1\,\dot{M}_{\rm Edd}$ thick-disk threshold). Each cell shows the per-Eddington-normalized accretion rates of the primary (black) and secondary (red) on a log y-axis (range shifted per panel to match the data); the dashed gray horizontal marks the $\dot{M}_i = \dot{M}_{{\rm Edd},i}$ super-Eddington threshold and the dotted gray horizontal marks the $\dot{M}_i = 0.01\,\dot{M}_{{\rm Edd},i}$ ADAF threshold. Backgrounds are colored by jet regime: blue (single-jet, one BH meets its panel's threshold), purple (dual-jet, both BHs meet their threshold simultaneously sustained), green (flickering, both meet threshold individually but rarely simultaneously). At both extremes dual jets dominate; the fiducial $\dot{M}_b = 1\,\dot{M}_{\rm Edd}$ shown in Fig. 11 represents the radiatively-efficient transition band where the regime mix is richest.

Thus, we are left with the set of coupled differential equations

$$\dot{a}(e_b, q_b) = \dot{a}_{\rm gas}(e_b, q_b) + \dot{a}_{\rm GW}(q_b, e_b, a_b) \tag{12}$$
$$\dot{e}(e_b, q_b) = \dot{e}_{\rm gas}(e_b, q_b) + \dot{e}_{\rm GW}(q_b, e_b, a_b) \tag{13}$$
$$\dot{q}(e_b, q_b) = \dot{q}_{\rm gas}(e_b, q_b) \tag{14}$$
$$\tag{15}$$

which we numerically integrate given an initial a_0, q_0, and e_0.

Using $\langle\dot{q}\rangle$ Fig. 10, we solve the initial value problem for two equal-mass binaries with $e_b = 0.2$ and $e_b = 0.3$, integrating forward for $10^4\tau$ with $a_0 = 10^3 R_S$. The results are displayed in Fig. 15. We assume a binary accretion rate $\dot{M}_b = 100\,\dot{M}_{\rm Edd}$ throughout. This assumption sets the timescale over which the gas-driven equilibrium is reached but does not change the qualitative endpoint of the evolution over our integration interval. We retain time as the x-axis in Fig. 15 (rather than semi-major axis) because the relaxation to the gas-driven equilibrium is more naturally read in time units.

In Fig. 15 we display the evolution of the two binary's $e(t)$ and $q(t)$ in the upper and lower panels, respectively. Our numerical integration is performed at $a_0 = 10^3 R_S$, where the gas-driven terms $\dot{a}_{\rm gas}$ and $\dot{e}_{\rm gas}$ dominate over the GW terms (Peters 1964), and the binary settles toward the gas-driven equilibrium $e \approx 0.45$ reported by Duffell et al. 2020; Zrake et al. 2021; Siwek et al. 2023b, 2024. We find that the two binaries initialized at $(e_b, q_b) = (0.2, 1.0)$ and $(0.3, 1.0)$ both evolve away from $q_b = 1$, reaching $(e, q) = (0.45, 0.991)$ and $(0.45, 0.998)$ respectively, before further evolution.

These plateau values are not the final pre-merger state. As the binary inspirals and a shrinks, the GW timescale $\tau_{\rm GW} \propto a^4$ (Peters 1964) eventually becomes shorter than the gas-driven evolution timescale, and GW radiation begins to dominate the orbital dynamics. Crucially, the Peters quadrupole formulas conserve the mass ratio q at leading order: GW radiation drives $a \to 0$ and $e \to 0$ but does not directly change q. The mass-ratio offset established during the gas-dominated phase is therefore preserved through inspiral rather than driven back toward unity. We have verified this by extending the integration of equation 12 for the $(e_b, q_b) = (0.2, 1.0)$ and $(0.3, 1.0)$ cases from $a_0 = 10^3 R_S$ down to $a = 5R_S$: GW circularizes e to ≈ 0 before LISA-band entry, but q remains essentially constant at the gas-phase plateau. A scan over initial separation finds that the bump effect activates by $a_0 \gtrsim 250 R_S$ and reaches its asymptotic value by $a_0 \gtrsim 400 R_S$; for an $M_{\rm tot} = 10^7 M_\odot$ binary entering inspiral with $\dot{M}_b = 100\dot{M}_{\rm Edd}$, the asymptotic mass-ratio offsets at LISA-band entry are $\Delta q \approx 0.014$ and $\Delta q \approx 0.008$ for the two cases, respectively.

Two caveats apply to this prediction. First, the simulation suite that supplies $\dot{q}_{\rm gas}$ is gridded at $\Delta q_b = 0.1$ resolution. Once q drifts from 1.0 to ~0.99 during inspiral, our integration samples the $q_b = 0.9$ row of the lookup table, which corresponds to simulations of

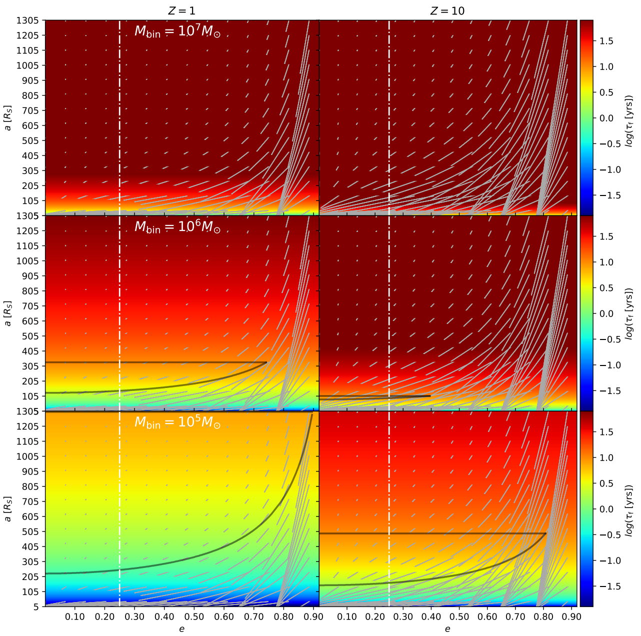

Figure 14. Region of (e_b, a_b) parameter space in which a system satisfies the flicker observability constraints (a flicker time $\tau_f \le 10$ yr in the observer frame, and a remaining merger time ≥ 100 yr) for various binary masses and redshifts. The black contours enclose the detection-friendly region in each panel. The x-axis of each panel is the binary eccentricity e_b, the y-axis is the binary semi-major axis a_b in Schwarzschild radii, and τ_f is reported in the observer frame. The gray lines depict the net change in (e_b, a_b) a binary undergoes over 10 orbits of GW emission.

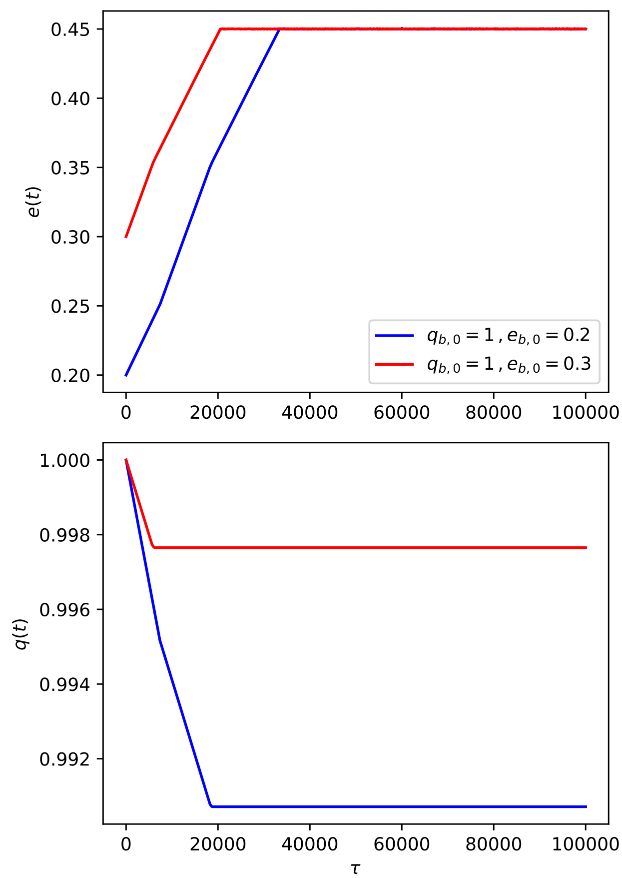

Figure 15. Evolution of binaries away from equal mass. We depict the result of our numerical integration of equation 12 for $(e_{b_0}, q_{b_0}) = (0.2, 1.0)$ (blue) and $(0.3, 1.0)$ (red). The y-axes are the eccentricity $e(t)$ (upper) and mass ratio $q(t)$ (lower) of the binary; the x-axes are time in units of the binary orbital period τ_b.

binaries that had $q_b = 0.9$ throughout — not binaries that started at $q = 1$ and drifted. We do not have direct simulation data in $q \in (0.95, 1.0)$. Second, the cavity geometry depends on the binary's history: a binary that started at $q = 1$ has a CBD locked toward the spatially-defined "secondary" (an apocenter convention), whereas a $q_b = 0.9$ simulation has a CBD locked toward the lower-mass BH from the start. As q drifts past unity, the cavity must reorient, which we argued in Section 3.2 proceeds on the disk's apsidal precession timescale — a quantity we have not compared to the gas-driven q-evolution timescale. If the cavity reorients faster than q evolves, the lookup remains applicable; if slower, the relevant $\dot{q}_{\rm gas}$ during inspiral could differ from what our integration assumes. Resolving these caveats would require new simulations at finer q-grid resolution and live-binary runs through a sustained $\dot{q} \neq 0$ phase, both of which we leave to future work.

Subject to these caveats, we predict that LISA-detectable binaries that passed through $(e_b, q_b) = (0.2, 1.0)$ or $(0.3, 1.0)$ during the gas-dominated phase of their formation should appear at $q \approx 0.99$ rather than at $q = 1$, producing a relative dearth of strictly $q = 1$ systems in the LISA mass-ratio distribution. Whether this leaves a measurable population-level signature depends on three additional open questions: (i) the fraction of MBHB progenitors that pass through the relevant region of (e_b, q_b) parameter space and the duty cycle of the $q \neq 1$ phase across the population, (ii) LISA's expected ~0.5% precision in q (Mangiagli et al. 2020) relative to the $\Delta q \sim 0.01$ deviation, and (iii) the expected $O(100)$ MBHB events over the LISA mission. Confirming this prediction would require both a careful population synthesis accounting for the duty cycle of the $q \neq 1$ phase and a forward-modelling of LISA's detection sensitivity for a near-equal-mass population.

5 SUMMARY AND CONCLUSIONS

This paper has provided the most extensive report to date on preferential accretion $\lambda \equiv \frac{\dot{M}_2}{\dot{M}_1}$ and mass ratio rate of change $\dot{q}$ for SMBBHs embedded in CBDs. In addition to calculating these quantities, we provide novel insight into the behavior of these quantities over time and their dependence on q_b and e_b. We also conduct a preliminary investigation into how the CBD regulates preferential accretion. We summarize our key findings below.

(i) Across q_b and e_b, $\lambda(t)$ can be split into constant and time-varying regimes (Table 1), broadly mirroring the split of the CBD into locked and precessing states.

(ii) The average preferential accretion value $\langle\lambda\rangle$ exhibits a weak positive correlation with both e_b and q_b (Fig. 5). The standard deviation of λ, σ_λ, exhibits a weak negative correlation with both e_b and q_b (Fig. 2).

(iii) $\langle\lambda\rangle$ is positively correlated with more eccentric CBDs (Fig. 2, compare Fig. 3).

(iv) Across precessing systems in our suite, the CBD apsidal precession period and the $\lambda(t)$ oscillation period are equal. However, these time-series are not in phase, ruling out a simple causal interpretation in terms of cavity-wall distance.

(v) We do not find evidence that a BH must be closer to the CBD cavity than its counterpart to accrete at a higher rate than its counterpart.

(vi) Normalized to Eddington accretion rates, $\lambda(t)$ results in disparate accretion regimes for each BH in the binary—leading to unique jet-launching regimes. We delineate binaries that are likely to launch a sustained jet from one BH (single-jet), from each BH (dual-jet), or alternate in which BH launches a jet (flickering-jet).

(vii) Equal-mass binaries at $e_b = 0.2$ and 0.3 are not long-lived equilibrium states: during the gas-dominated phase of evolution they preferentially accrete away from equal mass, reaching $q \approx 0.991$ and 0.998 respectively. Because GW radiation conserves the mass ratio at leading order, this offset is preserved through inspiral. We therefore predict a relative dearth of strictly $q = 1$ systems in the LISA mass-ratio distribution; whether this is detectable depends on population statistics and forward-modelling we leave to future work.

While our work has shed light on one aspect of the SMBBH–CBD system, we are still studying setups with simplified physics, namely two-dimensional hydrodynamic simulations. In future work we hope to expand this study to more physically motivated simulation setups, accounting for effects that were ignored in this study. Namely, we hope to detail λ and $\dot{q}$ for radiation MHD simulations in three dimensions. Further, we would like to also develop GRMHD simulations for SMBBHs in order to better simulate jet-launching and test our findings on different jet regimes. In addition to developing more robust simulations, we would also like to run live-binary simulations for the $(e_b, q_b) = (0.2, 1.0)$ and $(0.3, 1.0)$ binary systems to not only better extrapolate how the binary evolves away from unity but to also understand how the disk reacts to the change in position of the primary and secondary. Finally, we will further investigate the mechanism behind preferential accretion, with a forthcoming analysis on the hydrodynamics of the gas within the cavity.

We conclude by noting that the mechanism behind preferential accretion is more complex than previously thought. Although our results are consistent with the CBD playing a regulating role, the most concrete observational handle is the flickering-jet regime; the population-level LISA signature is intriguing but conditional on the binary formation distribution. We encourage further study into both.

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DATA AVAILABILITY

The data underlying this article will be shared on reasonable request to the corresponding author.

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